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AI/ML Use-Case Decision Framework for Financial Services

A practical reference for matching data characteristics to the right technique.

Purpose

One of the most common questions in applying AI/ML to a business problem isn't 'can we use AI here,' but 'which approach actually fits this problem.' This framework distills that judgment into four practical categories, each matched to a data pattern, a recommended technique, and a real finance example, so the right starting point can be identified quickly before deeper analysis begins.

1. No Labeled Outcomes: Unsupervised Learning

When there is no predefined label to predict and the goal is to discover hidden structure in the data itself, clustering-based unsupervised learning is the right starting point. This applies directly to customer segmentation, where a bank might group customers by behavior without predefined categories, and to transaction anomaly detection, where the model flags unusual activity without ever being told what 'normal' looks like in advance.

2. Structured, Labeled Data with Limited Features: Traditional Machine Learning

When historical outcomes are known (churned or not, defaulted or not) and the relevant features are well understood and limited in number, traditional ML algorithms such as Support Vector Machines, Logistic Regression, or Decision Trees are typically the better choice. These models train efficiently on smaller, structured datasets and produce interpretable results, which matters when a bank needs to explain why a customer was flagged for churn risk or denied credit. Common applications include customer churn prediction and credit risk scoring.

3. Raw, High-Volume, Sequential Data: Deep Learning

When the data is unstructured, high-dimensional, or arrives as a continuous sequence, and manually engineering features would be impractical, deep learning architectures such as RNNs, CNNs, or Transformer models become necessary. These models automatically learn hierarchical patterns directly from raw data. In finance, this shows up in real-time fraud detection, where RNNs track sequences of transactions to flag anomalies as they happen, and in contract or document analysis, where Transformer-based models process unstructured text at scale.

4. Sequential Decisions with Reward-Based Feedback: Reinforcement Learning

When a problem involves a sequence of decisions that are evaluated over time through reward and penalty signals, rather than a single prediction against a known label, reinforcement learning is the appropriate technique. Algorithmic trading strategy optimization is the clearest financial example: a model takes a sequence of trading actions and adjusts its strategy based on the resulting financial outcomes, gradually improving performance through continuous feedback.

How to Use This Framework

Start by asking two questions: first, is there a labeled outcome to predict, or is the goal to discover unknown structure? Second, is the data structured and limited, or raw, high-volume, and continuous? The answers to these two questions point toward one of the four categories above, which then serves as a starting hypothesis for deeper technical evaluation, not a final answer. This mirrors the reasoning applied in my ML vs. Deep Learning case study report, which walks through two of these categories in full depth using real financial services examples.